

Sam Altman is wrong: humans are much more energy efficient than AI data centers

Posted by u/SkankHuntThreeFiddy • 0 points • 66 comments

The average daily energy consumption of a human is 2000 Calories (kcal), or about 2.326 kilowatt-hours (kWh), per day. Energy of 2.326 kWh consumed evenly over 24 hours is equal to 97 watts of power. Humans exhale carbon dioxide, which has a global warming potential of 1.

The average 1U rack-mounted server uses 800 watts of power. Over 24 hours, that's 19.2 kWh, or about 16,500 kcal, of energy. A common refrigerant used in air conditioners in data centers is difluoromethane, which has a global warming potential of 677.

Put simply: a single computer in a data center uses as much energy as 8.25 humans, and its emissions contribute to global warming 677 times more than a human exhaling.

Comments

u/Cata135 • 1 points • 2026-03-04 23:25 UTC

Yeah, that doesn't prove your use case either. If a single computer does more than 8-9x more work, then it would still be more energy efficient than the human. For things like generating images, computers are much more energy efficient/time efficient/resource efficient than humans.

u/GurSlight • 1 points • 2026-02-28 01:17 UTC

Even though we exhale carbon dioxide of 1, you didn't take account on the emissions of the food that we consume. That food is harvest by machinery, transported to a processing facility, then transported to a grocery store. Then you drive to the grocery store.

We need to take that inconsideration if we want a apples to apples comparison

u/Unfair_Battle5564 • 1 points • 2026-02-26 16:01 UTC

He is a crazy person hack forgetting he isn't supposed to say "we think all non billionaires should be exterminated and their labor replaced with robots"

u/Latter-Corner8977 • 1 points • 2026-02-26 12:30 UTC

Really hate him for spouting shite like this. Genuinely. Traitor to his fucking species.

u/LuminaUI • 1 points • 2026-02-26 02:08 UTC

That doesn't include the entire economic system required to sustain that human like agriculture, transport, housing, heating/cooling, infrastructure, etc.

So the 97W number is biologically true, but that's your base metabolic rate.

That's only gonna be close to true when the AI puts us all in jars to use as batteries.

u/DueCommunication9248 • 0 points • 2026-02-25 20:09 UTC

Didn't even mention where the energy for humans comes from!?

We're not powered by electricity, those 2000kcal have to be formed and grown for months to decades.

u/konglongjiqiche • 1 points • 2026-02-25 13:08 UTC

The refrigerant is recycled, it's not a side effect of the energy consumption. Do you mean eventually it's released when the cooling system is disposed?

u/tomvorlostriddle • 1 points • 2026-02-25 08:55 UTC

I know that the sub is called better offline, but living like a literal hunter gatherer tribe, which is implied by that 200kCal lifestyle, might be a bit extreme

\> Put simply: a single computer in a data center uses as much energy as 8.25 humans

And how many does it serve?

u/sandyflame • 1 points • 2026-02-25 07:16 UTC

you are forgetting the cost of humans ...we don't do anything for nothing. we want reasonable pay, holidays sick days etc etc,.....and we are not really productiveeven with all of that

u/sparqq • 1 points • 2026-02-25 02:38 UTC

That Sam Altman is getting as popular as a healthcare CEO

u/Budz_Buddha • 1 points • 2026-02-24 23:25 UTC

I mean, he's wrong, but as an hvac guy, you're wrong, too. It's a closed loop system. All HCFC (Hydrochlorofluorocarbon) and cfc refrigerants are awful for the ozone. The entire premise of a hvac system is phase change using the pressure, temperature, and the unique boiling point of refrigerant.

It's nuance but it has a potential to be 677x worse than a human if it were to develop a leak.

u/Such_Degree3034 • 1 points • 2026-02-24 21:51 UTC

Humans are part of the nature energy cycle, what humans get from nature they return to nature, data centers do not.

u/Old-Artist-5369 • 1 points • 2026-02-24 19:35 UTC

I don't agree with Altman's point. But you've missed it so badly it's embarrassing.

The comparison on the refrigerant is also stupid.

u/ChxsenK • 1 points • 2026-02-24 13:30 UTC

Of course he is wrong. Takes 20 years under this bullshit education system, designed to dumb down, indebt and gatekeep free thinking and creativity.

u/duboispourliver • 1 points • 2026-02-24 13:03 UTC

Maths are applied wrong here, and the 1U server cost is not what's important.

What's important is that with the 2kWh the human has consumed, you can power around 2000 GPT requests. Does a human produce more in a day than 2000 GPT requests? That's the relevant approach.

But there's more. A human releases 2kWh per day of energy, ok. But you don't plus humans on electric wires. You have to produce food for humans, and bring entertainment, etc. The total energy cost of a human is not about his metabolic energy, it's about all that. I think you can factor by 20 at least.